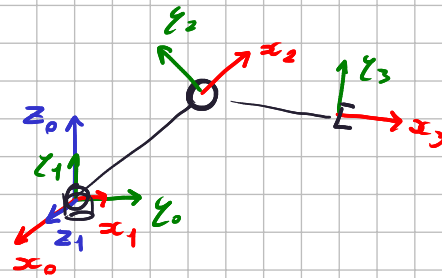


Ex 1) First we write the DH transformations.

$${}^0T_1 = \begin{pmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1T_2 = \begin{pmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^2T_3 = \begin{pmatrix} c_3 & -s_3 & 0 & a_3 c_3 \\ s_3 & c_3 & 0 & a_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Now we should write the centers of mass positions.

$$\begin{cases} {}^1r_{c1} = (0 \ 0 \ 0)^T & \text{the first link have zero length.} \\ {}^2r_{c2} = (-\frac{a_2}{2} \ 0 \ 0)^T \\ {}^3r_{c3} = (-\frac{a_3}{2} \ 0 \ 0)^T \end{cases}$$



Now we apply the algorithm:

$${}^1W_1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} \quad {}^1v_1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix} \left[\begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right] = 0$$

$$\Rightarrow {}^1v_{c1} = 0 \Rightarrow T_1 = \frac{1}{2} I_{1x} \dot{q}_1^2$$

$${}^2W_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ \dot{q}_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} \right] = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_1 \end{pmatrix} = \begin{pmatrix} \dot{q}_1 s_2 \\ \dot{q}_1 c_2 \\ \dot{q}_2 \end{pmatrix}$$

$${}^2v_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_1 \end{pmatrix} \times \begin{pmatrix} a_2 c_2 \\ a_2 s_2 \\ 0 \end{pmatrix} \right] = \begin{pmatrix} 0 \\ a_2 \dot{q}_2 \\ -a_2 \dot{q}_1 c_2 \end{pmatrix}$$

$${}^2v_{c2} = {}^2v_2 + {}^2W_2 \times {}^2r_{c2} = \begin{cases} 0 \\ \frac{1}{2} a_2 \dot{q}_2 \\ -\frac{1}{2} a_2 \dot{q}_1 c_2 \end{cases} \Rightarrow T_2 = \frac{1}{2} m_2 \left[\frac{1}{4} a_2^2 \dot{q}_2^2 + \frac{1}{4} a_2^2 c_2^2 \dot{q}_1^2 \right] + \frac{1}{2} \left[I_{2x} \dot{q}_2^2 + \dot{q}_1^2 s_2^2 I_{2x} + \dot{q}_1^2 c_2^2 I_{2y} \right]$$

$$T_2 = \frac{1}{2} \left[\dot{q}_1^2 \left(c_2^2 \left[m_2 a_2^2 \frac{1}{4} + I_{2y} \right] + s_2^2 I_{2x} \right) + \dot{q}_2^2 \left(\frac{1}{4} m_2 a_2^2 + I_{2x} \right) \right]$$

$${}^3W_3 = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \dot{q}_1 s_2 \\ \dot{q}_1 c_2 \\ \dot{q}_2 + \dot{q}_3 \end{pmatrix} = \begin{cases} \dot{q}_1 s_{23} \\ \dot{q}_1 c_{23} \\ \dot{q}_2 + \dot{q}_3 \end{cases}$$

$${}^3v_3 = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ a_2 \dot{q}_2 \\ -a_2 \dot{q}_1 c_2 \end{pmatrix} + \begin{pmatrix} \dot{q}_1 s_2 \\ \dot{q}_1 c_2 \\ \dot{q}_2 + \dot{q}_3 \end{pmatrix} \times \begin{pmatrix} a_3 c_3 \\ a_3 s_3 \\ 0 \end{pmatrix} \right] = \begin{cases} a_2 \dot{q}_2 s_3 \\ \dot{q}_2 (a_3 + a_2 c_3) + a_3 \dot{q}_3 \\ -\dot{q}_1 (a_3 c_{23} + a_2 c_2) \end{cases}$$

$$v_{c3} = \begin{cases} a_2 \dot{q}_2 s_3 \\ \frac{1}{2} a_3 \dot{q}_2 + \frac{1}{2} a_3 \dot{q}_3 + a_2 \dot{q}_2 c_3 \\ -\frac{1}{2} a_3 \dot{q}_1 c_{23} - a_2 \dot{q}_1 c_2 \end{cases} = \begin{cases} a_2 \dot{q}_2 s_3 \\ \frac{1}{2} a_3 (\dot{q}_2 + \dot{q}_3) + a_2 \dot{q}_2 c_3 \\ -\dot{q}_1 (a_3 c_{23} + a_2 c_2) \end{cases}$$

$$T_3 = \frac{1}{2} m_3 [\dot{a}_2^2 \dot{q}_2^2 + \frac{1}{4} a_2^2 (\dot{q}_2^2 + \dot{q}_3^2 + 2\dot{q}_2 \dot{q}_3) + a_2 a_3 c_3 (\dot{q}_2^2 + \dot{q}_2 \dot{q}_3) + \dot{q}_1^2 (a_3^2 c_{23}^2 + a_2^2 c_2^2 + 2a_2 a_3 c_2 c_{23})] + \frac{1}{2} [\dot{q}_1^2 (s_{23}^2 (I_{3x} - I_{3y}) + I_{3z}) + I_{3z} (\dot{q}_2^2 + \dot{q}_3^2 + \dot{q}_2 \dot{q}_3)]$$

$$M_{11} = I_{1y} + c_2^2 [m_2 a_2^2 \frac{1}{4} + I_{2y} - I_{2x}] + I_{2x} + m_3 (a_3^2 c_{23}^2 + a_2^2 c_2^2 + 2a_2 a_3 c_2 c_{23}) + (c_{23}^2 (I_{3y} - I_{3x}) + I_{3x})$$

$$c_2^2 [m_2 a_2^2 \frac{1}{4} + I_{2y} - I_{2x} + m_3 a_2^2] + c_{23}^2 [m_3 a_3 + I_{3y} - I_{3x}] + m_3 c_2 c_{23} [2a_2 a_3] + I_{1y} + I_{2x} + I_{3x} = e_1 + e_2 c_2^2 + e_3 c_{23}^2 + 2e_4 c_2 c_{23}$$

$$M_{22} = \frac{1}{4} m_2 a_2^2 + I_{2x} + m_3 (a_2^2 + \frac{1}{4} a_3^2) + m_3 a_2 a_3 c_3 + I_{3z} = e_5 + e_4 c_3$$

$$M_{23} = \frac{1}{4} m_3 a_3^2 + \frac{1}{2} m_3 a_2 a_3 c_3 + I_{3z} = \frac{1}{2} e_4 c_3 + e_6$$

$$M_{33} = I_{3z} + \frac{1}{4} m_3 a_3^2 = e_6$$

$$M = \begin{bmatrix} e_1 + e_2 c_2^2 + e_3 c_{23}^2 + 2e_4 c_2 c_{23} & 0 & 0 \\ 0 & e_5 + e_4 c_3 & \frac{1}{2} e_4 c_3 + e_6 \\ 0 & \frac{1}{2} e_4 c_3 + e_6 & e_6 \end{bmatrix}$$

$$M\ddot{q} = \begin{bmatrix} \ddot{q}_1 & c_2^2 \ddot{q}_1 & \ddot{q}_1 c_{23}^2 & 2c_2 c_{23} \ddot{q}_1 & 0 & 0 \\ 0 & 0 & 0 & c_3 \ddot{q}_2 + \frac{1}{2} c_3 \ddot{q}_3 & \ddot{q}_2 & \ddot{q}_3 \\ 0 & 0 & 0 & 0 & \frac{1}{2} c_3 \ddot{q}_2 & \ddot{q}_2 + \ddot{q}_3 \end{bmatrix} e$$

Ex 2) We can control $\dot{q}$ and we want to impose a desired angle α_d where $\alpha = \sum q_i \Rightarrow J = (1 \ 1 \ 1 \ 1)$

$$\dot{\alpha} = J \dot{q} \text{ we impose } \dot{\alpha} = \underbrace{(\alpha_d - \sum_i q_i)}_{\text{Feedback}} + \underbrace{\dot{\alpha}_d}_{\text{Feedforward}}$$

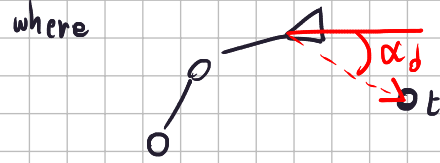
so $\dot{q} = J^\# \dot{\alpha}$ but we want also to move the joints to the midrange. let $G = \frac{1}{2} (q_1^2 + q_3^2 + q_4^2 + (\frac{\pi}{4} - q_2)^2)$

$$\nabla G = (q_1 \quad -2(\frac{\pi}{4} - q_2) \quad q_3 \quad q_4)^T \quad \text{We set}$$

$$\dot{q} = J^\# \left[(\alpha_d - \sum_i q_i) + \dot{\alpha}_d \right] - \underbrace{(\mathbf{I} - J^\# J)}_{\text{project on null space}} \nabla G$$

What is α_d ? is $\arctan \left\{ \frac{y - p_y}{x - p_x} \right\}$

$(p_x, p_y) = \text{end eff.}$ $(x, y) = \text{target}$



here the numerical calculations are skipped.

Ex 3) We proceed with a $\mathcal{L}_{\text{Lapunov}}$ candidate, based on the energy

$$V = \frac{1}{2} \dot{q}^T M \dot{q} + \frac{1}{2} \dot{\theta}^T B_m \dot{\theta} + \frac{1}{2} (q - \theta)^T K (q - \theta) + \underbrace{(\theta_d - \theta)^T K_p (\theta_d - \theta)}_{\mathcal{L}_{\text{virtual spring}}}$$

This is zero when $\dot{\theta} = \dot{q} = 0$, $q = \theta = \theta_d$. Now we compute the derivative

$$\dot{V} = \dot{q}^T \dot{M} \dot{q} + \dot{q}^T M \ddot{q} + \dot{\theta}^T B_m \ddot{\theta} + (q - \theta)^T K (\dot{q} - \dot{\theta}) - (\theta_d - \theta)^T K_p \dot{\theta} \Rightarrow$$

$$\dot{V} = \dot{q}^T (\underbrace{\dot{M} \dot{q}}_{\text{cancels}} - C - K(q - \theta)) + \dot{\theta}^T (K_p (\theta_d - \theta) - K_d \dot{\theta} - K(\theta - q)) + (q - \theta)^T K (\dot{q} - \dot{\theta}) - (\theta_d - \theta)^T K_p \dot{\theta}$$

$$= \dot{q}^T K (q - \theta) + \dot{\theta}^T K (\theta - q) + \dot{\theta}^T (K_p (\theta_d - \theta) - K_d \dot{\theta}) - (q - \theta)^T K (\dot{q} - \dot{\theta}) - (\theta_d - \theta)^T K_p \dot{\theta}$$

$$\text{Notice: } \dot{q}^T K (q - \theta) + \dot{\theta}^T K (\theta - q) - (q - \theta)^T K (\dot{q} - \dot{\theta}) = 0$$

$$\dot{V} = \dot{\theta}^T (K_p (\theta_d - \theta) - K_d \dot{\theta}) - (\theta_d - \theta)^T K_p \dot{\theta} = -\dot{\theta}^T K_d \dot{\theta} < 0$$

so $\dot{V} < 0 \Rightarrow$ the system converge to the largest inv. set. Now we

consider $\dot{V} = 0 \Rightarrow \dot{\theta} = 0$

$$\begin{cases} K(q - \theta) = 0 \\ K(\theta - q) = K_p(\theta_d - \theta) \end{cases} \begin{cases} q = \theta \\ \theta_d = \theta \end{cases} \Rightarrow \text{the system converge to } \begin{cases} \theta = \theta_d \\ q = \theta_d \\ \dot{\theta} = 0 \\ \dot{q} = 0 \end{cases}$$

Ex 4) The general residual method is defined as follows:

$r = K[-p + \int (z - \alpha - r) dt]$, we are interested only on r_1 so we compute: $\alpha_1 = -\frac{1}{2} \dot{q}^T \frac{\partial M}{\partial \dot{q}_1} \dot{q} + g_1$, We need M and g

$$p_{c1} = \begin{cases} 0 \\ q_1 \end{cases} \quad p_{c2} = \begin{cases} d_{c2} \dot{q}_2 \\ q_1 - d_{c2} s_2 \end{cases} \quad \dot{p}_{c1} = \begin{cases} 0 \\ \dot{q}_1 \end{cases} \quad \dot{p}_{c2} = \begin{cases} -d_{c2} s_2 \dot{q}_2 \\ \dot{q}_1 - d_{c2} \dot{q}_2 \end{cases}$$

$$U = -g_0 (m_1 q_1 + m_2 q_1 - m_2 d_{c2} s_2) \Rightarrow g_1 = -g_0 (m_1 + m_2) = -g_0 e_1$$

$$T = \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} I_2 \dot{q}_2^2 + \frac{1}{2} m_2 (d_{c2}^2 \dot{q}_2^2 + \dot{q}_1^2 - 2 d_{c2} \dot{q}_1 \dot{q}_2)$$

$$M_{11} = m_1 + m_2 \quad M_{12} = -m_2 d_{c2} \quad M_{22} = I_2 + m_2 d_{c2}^2$$

$\begin{matrix} e_1 & & e_3 \\ & -e_2 c_2 & \end{matrix}$

$$M = \begin{pmatrix} e_1 & -e_2 c_2 \\ -e_2 c_2 & e_3 \end{pmatrix} \quad \frac{\partial M}{\partial \dot{q}_1} = 0 \text{ (trivial!)} \Rightarrow \alpha_1 = -g_0 e_1$$

$$p_1 = (e_1 - e_2 c_2) \dot{q} = e_1 \dot{q}_1 - e_2 c_2 \dot{q}_2 \text{ hence}$$

$$r_1 = K_1 \left[\int (z_1 + g_0 e_1 - r) dt + e_2 c_2 \dot{q}_2 - e_1 \dot{q}_1 \right] \text{ this yields to}$$

$$\dot{r}_1 = K_1 (u_{s1} - r). \text{ Assume } u_{s1} = \begin{cases} 2 & t \geq 0 \\ 0 & t < 0 \end{cases} \text{ so we express in}$$

the Laplace domain

$$r_1 s = K_1 \left(\frac{2}{s} - r \right) \Rightarrow r_1 (s + K_1) = \frac{2K_1}{s} \Rightarrow r_1 = \frac{\frac{2K_1}{s}}{s + K_1} \text{ so}$$

$$\lim_{t \rightarrow \infty} r_1(t) = \lim_{s \rightarrow 0} s r_1(s) = \lim_{s \rightarrow 0} \frac{2K_1}{s + K_1} = 2$$

So r_1 will go exponentially to 2 with a transient defined by K_1